Adversary Trading Rules, Execution Policies & Economic Assumptions

Phase II — Adversarial PnL Leakage in Algorithmic Execution

Bocconi Students for Machine Learning (BSML) — Spring 2026

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Part A — Candidate Adversary Trading Rules & Execution Policies

Phase I Review — Key Takeaways

Phase I established the following baseline that Phase II builds upon:

- The execution desk runs a TSMOM (time-series momentum) strategy, splitting parent orders into child orders placed across intraday bars on a panel of liquid ETFs.
- A **passive** adversary predicted short-horizon returns using features derived from the child-order schedule but *did not trade*. Evaluation used information-theoretic metrics (AUC, MAE).
- **OU randomisation** of child-order placement timing was the only defence tested. It reduced adversarial predictability and improved implementation shortfall relative to a deterministic schedule.
- The simulator operated on a single execution policy with a single adversary; there was no PnL computation for either party.

Gaps to Close in Phase II

1. The adversary must become an **active trader** with realised PnL.
2. We need **multiple execution policies** (not just OU randomisation) to compare.
3. All results must be expressed in **basis points and annualised PnL**, not information metrics.
4. The simulator must support **multiple adversaries** $\times$ **multiple policies** in a single campaign.

Candidate Adversary Trading Rules

We propose two adversary archetypes. Each uses the same underlying signal (predicted short-horizon return from the Phase I ML models) but differs in *how* it trades on that signal. This ensures our policy conclusions are not artefacts of a single attacker model.

Archetype 1 — Aggressive Liquidity Taker (ALT)

Intuition

A high-frequency proprietary trader who detects our child-order flow and trades *directionally ahead* of our anticipated price impact, using market orders for immediate execution. The ALT aims to capture the short-term price move caused by our execution and unwinds promptly after.

Signal

Let $\hat{r}_{t,i}$ denote the adversary's predicted return for instrument i over the next execution interval $[t, t + \Delta]$, produced by the Phase I short-horizon model. The ALT trades when the predicted return exceeds a minimum threshold τ_{entry}.

Trading Rules

1. **Entry condition.** At the start of each execution interval t, if $|\hat{r}_{t,i}| > \tau_{\text{entry}}$:
   - If $\hat{r}_{t,i} > \tau_{\text{entry}}$ (we are expected to buy and push price up): the ALT **buys** q_{ALT} shares via a **market order** before our child order executes.
   - If $\hat{r}_{t,i} < -\tau_{\text{entry}}$: the ALT **sells** q_{ALT} shares.
2. **Position sizing.** $q_{\text{ALT}} = \min(\alpha_{\text{ALT}} \cdot V_{\text{bar},i},\ Q_{\max,i})$, where:
   - α_{ALT} is the participation-rate parameter (candidate values: 1%, 3%, 5% of bar volume),
   - $V_{\text{bar},i}$ is the average bar volume for instrument i,
   - $Q_{\max,i}$ is the per-instrument position limit (see Risk Controls below).
3. **Unwinding.** The ALT closes its position at the *end* of the same execution interval (or the next bar at latest) via a market order, capturing the price move during our execution window.
4. **No carry.** The ALT is flat by the end of each execution interval; overnight positions are forbidden.

Risk Controls

Parameter	Candidate Values	Description
$Q_{\max,i}$	5%, 10% of ADV	Maximum position per instrument as a fraction of average daily volume
L_{stop}	-5 bps, -10 bps	Per-trade stop-loss: if unrealised PnL hits this threshold, force immediate close
Π_{daily}	-50 bps	Daily drawdown limit: halt trading for the rest of the day if cumulative PnL breaches this level
Leverage	$1\times$ (no leverage)	The ALT trades with own capital only

Table 1: ALT risk-control parameters and candidate values.

Archetype 2 — Semi-Passive Spread Trader (SPST)

Intuition

A market-maker style adversary who leans on the informational edge from predicting our flow, but trades *passively* by posting limit orders on the side of our anticipated impact. The SPST earns the bid–ask spread while benefiting from a directional lean. This archetype is more realistic for an institutional market-maker and produces lower, steadier PnL than the ALT.

Signal

The same predicted return $\hat{r}_{t,i}$ from the Phase I model. The SPST also uses a *confidence threshold* τ_{lean} to decide how aggressively to skew its quotes.

Trading Rules

1. **Quoting.** The SPST continuously posts a two-sided quote (bid and ask) with a base half-spread of s_{base} bps around the mid-price.
2. **Directional lean.** When $\hat{r}_{t,i} > \tau_{\text{lean}}$ (expecting our buy flow to push the price up):

- **Tighten the ask** by δ_{lean} bps (making it easier for our buy child orders to lift the offer), and
- **Widen the bid** by δ_{lean} bps (reducing the chance of getting filled on the wrong side).

The reverse applies when $\hat{r}_{t,i} < -\tau_{\text{lean}}$.

3. **Inventory management.** If the SPST's net inventory $I_{t,i}$ exceeds a threshold $I_{\max,i}$, it cancels quotes on the accumulating side and posts a crossing order to reduce inventory.
4. **End-of-day flatten.** Any residual inventory at the close of the execution window is liquidated at the closing mid-price (simulating a guaranteed close auction).

Risk Controls

Parameter	Candidate Values	Description
s_{base}	1 bps, 2 bps, 3 bps	Base half-spread; wider = safer but less fill rate
δ_{lean}	0.5 bps, 1 bps	Lean adjustment magnitude
$I_{\max,i}$	2%, 5% of ADV	Maximum inventory per instrument before forced reduction
L_{stop}	-3 bps, -5 bps	Per-instrument intraday stop-loss
Leverage	$1\times$	No leverage

Table 2: SPST risk-control parameters and candidate values.

Adversary Summary Comparison

Characteristic	ALT	SPST
Order type	Market orders	Limit orders
Holding period	Within single bar	Within single bar
PnL profile	Lumpy, high variance	Steady, low variance
Turnover	High	Moderate
Information edge	Directional prediction	Directional lean + spread
Risk controls	Position limit + stop-loss	Inventory limit + stop-loss
Realistic analogue	Prop desk / HFT	Electronic market-maker

Table 3: Side-by-side comparison of the two adversary archetypes.

Candidate Execution Policies

We extend the Phase I policy space along three dimensions: randomisation type, aggressiveness, and adaptivity.

Dimension 1 — Randomisation Type

All three methods perturb the *timing* of child-order placement within a fixed execution horizon $[0, T]$. Let t_k^* denote the scheduled (deterministic) placement time of child order k, and t_k the actual (randomised) placement time.

Policy R1 — Ornstein–Uhlenbeck (OU) Randomisation

The perturbation follows a mean-reverting OU process:

$$d\epsilon_t = -\theta\, \epsilon_t\, dt + \sigma_{\mathrm{OU}}\, dW_t$$

where $\theta > 0$ is the mean-reversion speed and σ_{OU} is the volatility of the perturbation. The randomised placement time is $t_k = t_k^* + \epsilon_{t_k^*}$, clipped to $[0, T]$.

- **Candidate parameters:** $\theta \in \{0.5, 1.0, 2.0\}$; $\sigma_{\mathrm{OU}} \in \{0.01, 0.05, 0.10\}$ (as fraction of horizon T).
- **Properties:** Perturbations stay clustered around the schedule (mean-reverting), producing temporally correlated noise that is harder for the adversary to filter out than i.i.d. noise.
- **Phase I baseline:** This is the only randomisation tested in Phase I. We retain it as the reference policy.

Policy R2 — Uniform Randomisation

Each child order is placed at a uniformly random time within a window $[t_k^* - w,\ t_k^* + w]$:

$$t_k = t_k^* + U_k, \qquad U_k \sim \mathrm{Uniform}(-w,\, w)$$

clipped to $[0, T]$.

- **Candidate parameters:** Window half-width $w \in \{1, 3, 5\}$ bars.
- **Properties:** Perturbations are i.i.d. and bounded. Simple to implement and explain. The adversary can estimate the window width but not the exact realisation.

Policy R3 — Pink-Noise (1/f) Randomisation

The perturbation sequence $\{\epsilon_k\}$ is drawn from a $1/f$ (pink-noise) spectrum, producing long-range temporal correlations:

$$S(f) \propto \frac{1}{f^{\beta}}, \qquad \beta \approx 1$$

Generated via spectral synthesis: draw white-noise Fourier coefficients, scale amplitudes by $1/\sqrt{f}$, and inverse-FFT to obtain the time-domain perturbation series. The series is then scaled to a target standard deviation σ_{pink} and added to the deterministic schedule.

- **Candidate parameters:** $\sigma_{\mathrm{pink}} \in \{0.01, 0.05, 0.10\}$ (fraction of T); spectral exponent $\beta \in \{0.8, 1.0, 1.2\}$.
- **Properties:** Intermediate between OU (short-memory) and uniform (no memory). The long-range correlations may be harder for ML-based adversaries to learn because the perturbation "looks structured" but is not periodic.
- **Hypothesis to test:** Pink-noise randomisation may offer a better leakage/cost trade-off than OU in regimes where the adversary uses sequence models (RNNs, transformers) that exploit short-range autocorrelation.

Dimension 2 — Aggressiveness

Aggressiveness determines the *order type mix* and *participation rate* of our child orders. It affects both our execution cost and how visible our flow is to the adversary.

Level	Market Order %	Limit Order %	Participation Rate	Description
A1 — Passive	10%	90%	3–5% of bar volume	Mostly post limit orders; slow, low visibility, higher fill risk
A2 — Neutral	40%	60%	8–12%	Balanced mix; moderate cost and visibility
A3 — Aggressive	80%	20%	15–25%	Mostly take liquidity; fast completion, high impact, very visible

Table 4: Aggressiveness levels for our execution desk.

- **Limit-order modelling:** Limit orders are placed δ_{limit} bps inside the spread and are filled if the price touches or crosses the limit within the bar. Unfilled orders are cancelled at bar close and resubmitted the next bar (or converted to market orders under Aggressive).
- **Participation rate:** Caps the maximum fraction of bar volume our child orders can consume, preventing unrealistic market impact.

Dimension 3 — Adaptivity

- **Static policies (S):** All parameters (θ, σ, aggressiveness level) are fixed for the entire execution horizon. This is the Phase I approach.
- **Adaptive policies (D):** An escalation loop monitors rolling estimates of adversary PnL and our own implementation shortfall. When adversary PnL exceeds a threshold Π_{escalate} over a rolling window of W bars, the policy:
  1. Increases the randomisation intensity (e.g. $\sigma_{\text{OU}} \to \sigma_{\text{OU}} \cdot (1+\gamma)$).
  2. Optionally shifts aggressiveness toward Passive (more limit orders, lower participation).
  3. Logs the escalation event and new parameter values for post-hoc analysis.

  When adversary PnL drops below $\Pi_{\text{de-escalate}}$, the policy reverts toward baseline settings.

Adaptive-Loop Parameters

Parameter	Candidate Values	Description
W	10, 20, 50 bars	Rolling window length for adversary-PnL estimation
Π_{escalate}	2 bps, 5 bps	Adversary PnL threshold (per unit notional) that triggers escalation
$\Pi_{\text{de-escalate}}$	0.5 bps, 1 bps	Threshold for de-escalation back to baseline
γ	0.25, 0.50, 1.00	Multiplicative step-up for randomisation intensity
Aggressiveness shift	± 1 level	Optional: step from Aggressive $\to$ Neutral or Neutral $\to$ Passive on escalation

Table 5: Adaptive-loop candidate parameters.

Full Policy Grid

The complete combinatorial space is:

$$\underbrace{3}_{\text{Randomisation}} \times \underbrace{3}_{\text{Aggressiveness}} \times \underbrace{2}_{\text{Static / Adaptive}} = 18 \text{ policy configurations}$$

Each configuration is tested against both adversary archetypes (ALT and SPST), giving $18 \times 2 = 36$ experiment cells before regime stratification. In practice, we will run the full grid for a quick sweep, then select ∼6 "hero" policies for deep analysis with many Monte Carlo seeds.

Policy ID	Randomisation	Aggressiveness	Adaptivity
P1-S	OU ($\theta = 1.0$, $\sigma = 0.05$)	Neutral	Static
P2-S	Uniform ($w = 3$ bars)	Neutral	Static
P3-S	Pink ($\sigma = 0.05$, $\beta = 1.0$)	Neutral	Static
P4-S	OU ($\theta = 1.0$, $\sigma = 0.05$)	Passive	Static
P5-S	OU ($\theta = 1.0$, $\sigma = 0.05$)	Aggressive	Static
P1-D	OU ($\theta = 1.0$, $\sigma = 0.05$)	Neutral	Adaptive
...	*(remaining 12 combinations follow the same pattern)*		
Baseline	None (deterministic schedule)	Neutral	Static

Table 6: Representative subset of the policy grid. "Baseline" has no randomisation and serves as the reference for all comparisons.

Part B — Economic Assumptions & Policy-Space Validation

This section fulfils the Project Lead's Week 2 task: "Co-design economic assumptions behind the adversary (e.g. leverage limits, trading costs, risk appetite); review and validate the proposed policy space from a finance perspective."

Economic Assumptions — Our Execution Desk

The following assumptions anchor our desk's execution in realistic institutional terms.

Reference Notional and Instrument Universe

Assumption	Value	Rationale
Daily notional per instrument	USD 100 M	Round number representative of a mid-size systematic fund executing in liquid ETFs
Instrument universe	10–20 liquid US-listed ETFs	Continues Phase I panel; ensures sufficient volume for realistic fill assumptions
Execution horizon	Full trading day (6.5 hrs)	TSMOM signals are daily; execution spread across the session
Bar length	5 min or 15 min	Granularity of the intraday simulation; to be validated against Phase I data resolution

Transaction Costs — Our Desk

Cost Component	Assumption	Notes
Commission	0.5 bps per side	Institutional rate for liquid ETFs
Half-spread (market orders)	0.5–1.0 bps	Varies by ETF; calibrate from Phase I data
Market impact (temporary)	$\eta \cdot \left(\frac{q}{V_{\text{bar}}}\right)^{0.5}$	Square-root impact model; η calibrated per instrument
Market impact (permanent)	10–20% of temporary	Fraction that does not revert; conservative assumption
Limit-order fill probability	Empirical from Phase I	Conditional on distance from mid; unfilled orders carry opportunity cost

Benchmark Prices

- **Primary benchmark:** Arrival mid-price (mid at the time the parent order is generated). Implementation shortfall is measured relative to this price.
- **Secondary benchmark:** VWAP over the execution horizon. Useful for cross-checking and for clients who evaluate on a VWAP basis.

Economic Assumptions — Adversary

The adversary must operate under realistic constraints; otherwise leakage estimates will be inflated and conclusions will not transfer to practice.

Adversary Capital and Leverage

Parameter	Value	Rationale
Capital base	USD 10 M	Small relative to our USD 100 M daily flow; realistic for a prop desk or HFT pod
Leverage	1× (no leverage)	Simplifies PnL accounting and avoids inflated adversary returns
Margin / collateral	Not modelled	At 1× leverage, margin calls are not binding

Adversary Transaction Costs

The adversary faces the same fee schedule as our desk but may face *different* spread costs depending on its order type:

Cost Component	ALT	SPST	Notes
Commission	0.5 bps	0.3 bps	SPST may qualify for maker rebates
Half-spread paid	Full spread (taker)	Earns half-spread (maker)	ALT crosses the spread; SPST posts inside it
Market impact	Included via simulation	Negligible (small size)	ALT's impact is simulated; SPST posts passively

Adversary Risk Appetite

Constraint	Value	Finance Interpretation
Max position per instrument	5–10% of ADV	Prevents unrealistic concentration; consistent with a diversified prop book
Intraday stop-loss per trade	−5 to −10 bps	A disciplined desk cuts losers quickly
Daily drawdown limit	−50 bps of capital	If the adversary loses 50 bps on a USD 10 M book (−USD 5 000), it stops for the day
Overnight positions	Forbidden	Both archetypes must be flat by close; no carry risk

Finance perspective. These constraints ensure that the adversary PnL we report is achievable by a real-world participant. If we allowed unlimited leverage or no stop-losses, the adversary could generate arbitrarily large PnL in simulations, making leakage estimates meaningless for practitioner audiences.

Validation of the Policy Space

The Project Lead reviews the proposed policies from Part A through a finance lens, flagging issues and confirming suitability.

Randomisation Types — Finance Assessment

Verdict: All three randomisation types are approved for testing. Pink noise is flagged as experimental — if it underperforms or introduces unexplainable execution patterns, we can drop it by Week 4 without affecting the main conclusions.

Policy	Strengths	Risks / Concerns	Approved?
R1 (OU)	Mean-reverting; orders stay near schedule; proven in Phase I	The adversary may learn θ and partially de-noise the perturbation	✓
R2 (Uniform)	Simple; bounded perturbation; easy to explain to compliance	i.i.d. ⇒ adversary can average out noise over many observations	✓
R3 (Pink)	Long-range correlation may confuse sequence models	Harder to explain to a desk; may introduce autocorrelated execution delays	✓(test)

Level	Realistic for an Institutional Desk?	Key Trade-off
A1 (Passive)	Yes — common for low-urgency rebalancing	Lower impact cost, but higher fill risk and execution delay. The desk must tolerate partial fills.
A2 (Neutral)	Yes — default for most systematic funds	Balanced; the workhorse policy.
A3 (Aggressive)	Yes — used for urgent / high-alpha trades	Fast completion minimises signal decay but makes our flow maximally visible to the adversary.

Aggressiveness Levels — Finance Assessment

Verdict: The three-tier aggressiveness structure is realistic and well-calibrated. The candidate participation rates (3–5%, 8–12%, 15–25% of bar volume) are within typical institutional ranges for liquid ETFs.

Adaptive Loop — Finance Assessment

- **Practical concern:** A desk would only adopt an adaptive loop if it is *transparent and monitorable*. The escalation triggers (Π_{escalate}, $\Pi_{\text{de-escalate}}$) must be observable in real time and the parameter trajectories must be loggable.
- **Risk of over-fitting:** If the rolling window W is too short, the loop may oscillate between escalated and baseline modes, increasing turnover and cost. Recommend testing $W \geq 20$ bars as the default.
- **Desk narrative:** Frame the adaptive loop as analogous to a "risk thermostat" — it turns up defences when leakage heats up and dials them back in quiet periods. This framing is intuitive for portfolio managers.

Verdict: Approved for testing. Recommend adding a "cooldown period" of C bars after each escalation event during which no further changes occur, to prevent oscillation. Candidate values: $C \in \{5, 10, 20\}$ bars.

Metric Definitions (Confirmed)

The following metrics are agreed upon by the Quant Researcher and Project Lead for all experiments.

The Adversary Sharpe is included as a **plausibility check**: if an adversary archetype achieves a Sharpe ratio above ∼5 in simulation, its risk controls are likely too loose and should be tightened.

Metric	Unit	Definition
Implementation Shortfall (IS)	bps	$(P_{\text{avg,exec}} - P_{\text{arrival}})/P_{\text{arrival}} \times 10\,000$
Adversary PnL	bps / ann. USD	Mark-to-market of adversary round-trips, scaled to USD 100 M/day notional
Net Leakage	bps	Adversary PnL + excess IS (relative to a no-adversary baseline)
Fill Rate	%	Volume filled / volume intended, per child order
Execution Delay	bars (minutes)	Average lag between scheduled and actual fill time
Adversary Sharpe	—	Annualised adversary PnL / annualised adversary PnL std. dev. (sanity check)

Experiment Prioritisation — "Hero" Policies

For the deep-analysis phase (Weeks 4–6), we recommend focusing Monte Carlo resources on the following subset:

1. **Baseline** (no randomisation, Neutral aggressiveness, Static) — the reference.
2. **P1-S:** OU + Neutral + Static — Phase I default; continuity benchmark.
3. **P2-S:** Uniform + Neutral + Static — simplest alternative.
4. **P3-S:** Pink + Neutral + Static — experimental candidate.
5. **P4-S:** OU + Passive + Static — tests the cost of reducing visibility.
6. **P1-D:** OU + Neutral + Adaptive — tests whether the adaptive loop adds value over the best static policy.

Each hero policy will be run against both adversary archetypes, across all volatility/volume regimes, with $\geq$50 Monte Carlo seeds per cell.

Open Questions for Week 3

1. **Adversary signal lag.** Should the adversary observe our child orders with a 0-bar or 1-bar delay? A 0-bar delay is the worst case (adversary is co-located); 1-bar is more realistic for a non-HFT adversary.
2. **SPST fill model.** How do we model limit-order fills for the SPST? Options include a pro-rata model, a time-priority model, or a probabilistic fill function calibrated from Phase I data.
3. **Cross-instrument effects.** Should the adversary be allowed to use signals from instrument i to trade instrument j (e.g. correlated ETFs)? This is realistic but increases complexity.
4. **Pink-noise generation.** Confirm the spectral-synthesis implementation with the Simulation Engineer; ensure the generated series is stationary over the execution horizon.

This is a living design document. It will be updated as decisions are finalised in Weeks 3–4.